An Assessment of the Biases of Satellite Rainfall Estimates over the Tibetan Plateau and Correction Methods Based on Topographic Analysis

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ABSTRACT

The hydrological processes over the Tibetan Plateau have significant implications on regional macroscale atmospheric circulation patterns and the Asian monsoon system. Because of its remote setting and lack of ground observations, it is difficult to study the spatial and temporal patterns of precipitation over the plateau, and satellite remote sensing technology can be used to fill in the gaps where station data are not available. In this study the authors examine monthly $1° \times 1°$ rainfall estimates obtained from the Special Sensor Microwave Imager (SSM/I) [National Environmental Satellite, Data, and Information Service (NESDIS) algorithm] and Tropical Rainfall Measuring Mission (TRMM) 3B42 version 5 (V5) products for the months of April–October 1998–2002 over the Tibetan Plateau. By comparing the satellite estimates with ground observations at 94 weather stations in the study region, the authors derived regression models that produced significant improvements to satellite estimates based on various levels of correction efforts, using geographic location and topographic variables extracted from digital elevation models using geographic information systems (GIS) technology. The explained variance in observed precipitation was improved from 34% to 38% by SSM/I and TRMM 3B42 V5 products alone to over 70% when location and topographic variables were added. These topographic variables reflect micro- to mesoscale surface roughness, height of topographic features or relief, slopes facing toward or away from the moisture pathways, and relative locations or directions to prominent topographic features such as mountain peaks and ridgelines.

1. Introduction

The Tibetan Plateau has great significance in influencing regional atmospheric circulation patterns through its thermal and dynamic forcing mechanisms. With its vast size of approximately 2.3 million km^2, the plateau has a wide variety of environments, ranging from lush humid tropical montane forests in the southeastern part to highland arid lands and tundra in the interior and western parts (Zheng 1996). The plateau surface has elevations generally higher than 4000 m above mean sea level (MSL), with mountain ranges over 6000 m MSL. It is considered as an elevated heat source during the summer and a heat sink during the winter for the atmospheric processes in the region

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(Yeh and Gao 1979, 49–61). The hydrological cycle and energy balance processes on the plateau have far-reaching effects on the Asian monsoon systems (Ding 1991; Kutzbach et al. 1993; Ding and Chan 2005). The Tibetan Plateau is also the source region of several major rivers, including the Yangtze, Yellow, Lancang–Mekong, Salween–Nujiang, Ganges–Brahmaputra, and the Indus, which affect the livelihood of hundreds of millions of people in East and South Asia.

One important aspect of the investigations on the thermal forcing of the plateau on regional climate is the hydrological cycle, as a great amount of energy is involved in the latent heat flux of evapotranspiration and condensation processes, which influences the amount of sensible heat flux/warm air advection and the thermal gradient downwind (Ding and Chan 2005). There is a strong demand for accurate estimates of the components of hydrologic cycle, beginning with the amount of precipitation received over the plateau. Although there is an increasing interest in studying the changing environment of the plateau, due to a low density of weather stations, studies of climatic variability and hydrological processes over the Tibetan Plateau have been difficult. This is especially true because of high spatial and temporal variability of precipitation.

Since the 1980s, passive microwave remote sensing systems have been used to obtain rainfall estimates for regions with low weather station coverage. The Special Sensor Microwave Imager (SSM/I) aboard the Defense Meteorological Satellite Program (DMSP) satellites since 1987 and the Tropical Rainfall Measuring Mission (TRMM) Microwave Imager (TMI) aboard the TRMM satellite launched in 1997 are the two sensor systems that provide relatively long-term continuous estimates of precipitation worldwide for climatological applications (Ferraro et al. 1996; Grecu and Anagnostou 2001). For example, Negri et al. (2000) constructed a 10-yr climatology of Amazonian rainfall based on passive microwave satellite observations using the Goddard profiling algorithm. Besides applications in constructing precipitation climatology, satellite-based rainfall estimates have been used to study the spatial structure of individual storms. Among others, Nesbitt et al. (2004) compared the estimates from TMI and the precipitation radar (PR) of the TRMM satellite with other datasets at the global, regional, and storm scales. Additionally, Nesbitt and Zipser (2003) used the TRMM data to examine the diurnal cycle of rainfall events from small shallow features to large mesoscale convective systems. Barros et al. (2004, 2006) examined orographic precipitation using the TRMM data in the central Himalayas at micro-, meso-, and synoptic scales and identified topographic controls on cloud formation associated with the overall topography envelope and succession of ridges and valleys.

Previous studies have suggested that combining different sensors and platforms may improve the overall performance of satellite rainfall estimates. Additionally, such combinations would alleviate the problems of gaps in spatial coverage from nongeostationary satellites and low frequency of revisit, which reduce temporal resolutions especially in low-latitude regions. Bauer et al. (1998) explored the potential of joining TMI data with Visible Infrared Scanner (VIRS) data aboard the TRMM satellite. Adler et al. (2000) combined the TMI data with precipitation estimates based on the adjusted Geostationary Operational Environmental Satellite (GOES) precipitation index (AGPI), and then merged the satellite data with rain gauge data. Joyce et al. (2004) used the Climate Prediction Center morphing method (CMORPH) to combine the TMI data with the GOES infrared (IR) data to achieve high spatial and temporal resolutions in rainfall estimates, since passive microwave sensors in general have low spatial resolutions. Hong et al. (2004) used an artificial neutral network cloud classification system to improve precipitation estimates based on satellite IR images. Precipitation estimates from various sensor/platform sources were used in construction of the global gridded precipitation datasets, such as version 2 of the Global Precipitation Climatology Project (GPCP) monthly dataset (Adler et al. 2003) and the TRMM 3B43 datasets (Huffman et al. 2007). Although these gridded datasets incorporate station observations, for places with low station densities, such as over the Tibetan Plateau, the satellite estimates have the dominant influence.

Yao et al. (2001) demonstrated the feasibility of using TMI-derived brightness temperature to estimate precipitation with reasonable accuracy over the Tibetan Plateau. In an earlier study (Yin et al. 2004), it was found that the SSM/I-based rainfall estimates can be significantly improved by incorporating topographic and geographic location variables. The purpose of the current study is to improve satellite rainfall estimates over the Tibetan Plateau based on both SSM/I and TRMM data by incorporating topographic and location factors. Because of the similarities between the SSM/I and TRMM instruments (Wentz 1988; Grecu and Anagnostou 2001), we hypothesize that similar improvements should be seen for the TRMM data by including topographic and location variables in rainfall estimate algorithms. However, with different spatial coverage at any given time and different orbital characteristics, the two sensor systems may both contribute to the rainfall estimates with various levels of significance. We also want to investigate the seasonal varia-

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tion of the performance of the satellite estimates and topographic influences in satellite rainfall estimates. As regional atmospheric conditions and circulation patterns change seasonally, it is possible that the same set of topographic variables may have different effects in influencing the performance of the satellite rainfall estimates over time.

2. Data and methods

The SSM/I instrument has seven channels and four frequencies, measuring the atmospheric and surface microwave brightness temperatures at 19.35, 22.235, 37.0, and 85.5 GHz (Wentz 1988). The ground resolution (footprint) ranges from 13 to 69 km, depending on the channel and location along the 1394-km scanning swath. The TMI has the same channel/frequency setting and one added channel at 10.7 GHz to provide a more linear response for the high rainfall rates common in tropical rainfall (Grecu and Anagnostou 2001). The TMI also has an improved ground resolution because of a lower altitude (350 km as compared to 860 km of SSM/I). Additionally, there is a PR aboard the TRMM satellite. A wide variety of applications are developed based on the SSM/I and TRMM data, including rainfall, snow and ice, and soil moisture. The National Environmental Satellite, Data, and Information Service (NESDIS) algorithm of SSM/I rainfall rates over land is based on the scatter of microwave energy in the 85.5-GHz range by ice crystals and raindrop-sized cloud droplets in the entire atmospheric column (Ferraro et al. 1998). Although the footprint resolution of the SSM/I sensor ranges from 13 km at nadir to 69 km, gaps in observations during any given day in low-latitude regions as well as the twice-daily passing made total rainfall estimates reliable only for relatively long time periods. Monthly aggregation of all observations is often necessary to derive a relatively complete spatial coverage. The $1° \times 1°$ SSM/I monthly precipitation product from the National Oceanic and Atmospheric Administration (NOAA)/NESDIS was used in this study (Ferraro et al. 1998). The dataset is continuous from July 1987 to present, except for gaps in December 1987 and during July 1990 to December 1991.

The TRMM data used in this study consist of the $1° \times 1°$ 3B42 version 5 product from the National Aeronautics and Space Administration's (NASA) TRMM Online Visualization and Analysis System (TOVAS; http://daac.gsfc.nasa.gov/precipitation/TRMM_README/TRMM_README_V5/TRMM_3B42_readme.shtml and http://daac.gsfc.nasa.gov/precipitation/TRMM_README/TRMM_README_V5/TRMM_2B31_readme.shtml). We intentionally selected V5 of the dataset 3B42 for this analysis. In this dataset all footprints of the TMI and PR data of the TRMM satellite were first integrated (2B31 dataset) and then used to modulate the GOES IR data. The combined rain structure data of TRMM (version 5 of dataset 2B31) and VIRS calibration (1B01) were then used to estimate rainfall by calibrating the GOES precipitation index (GPI) data (Arkin et al. 1994). This approach allows filling the gaps in the TRMM measurements as the GOES data have a complete spatial and temporal coverage of the globe. The newer TRMM 3B42 version 6 (V6) is a much improved dataset. It uses a similar approach to combine the microwave data, including the TRMM, SSM/I, and Advanced Microwave Sounding Unit (AMSU) data (version 6 of 2B31), to modulate the GOES IR data (Huffman et al. 2007). For the comparison purposes, however, we used the V5 dataset, instead of the newer V6 dataset, because it does not have the effects of other satellite microwave rainfall estimates than just the TRMM estimates, and also because it has the same spatial resolution ($1° \times 1°$) as the SSM/I data. All rainfall estimates within the $1° \times 1°$ box were integrated for daily values and further summed to get the monthly values.

Our analysis focused on the area of 22°–40°N and 76°–108°E (Fig. 1). The spatial resolution of satellite data ($1° \times 1°$) and low-gauge density across the study region do not allow a detailed analysis on the physical processes of the topographic forcing mechanisms that cause biases in the satellite precipitation estimates. Therefore, we took an empirical approach to the examination of the spatial and temporal patterns of the biases in the satellite estimates in relation to many different measures of topographic features. In an earlier study (Yin et al. 2004), it was found that the SSM/I estimates performed poorly during the winter months because of the presence of snow and ice over the Tibetan Plateau, since snow and ice on the ground scatter microwave energy in a similar fashion as ice crystals and raindrops in the atmosphere. In the current study, we used data for the months of April through October of each year from 1998 to 2002 when both SSM/I and TRMM 3B42 V5 data, as well as the corresponding station data, were available. This period covers the monsoon season in which most of the rainfall for the year occurs. Monthly precipitation data for 94 weather stations were obtained from the Climatic Data Center of the China Meteorological Administration (Fig. 1; appendix A). The data have passed rigorous quality assessment and quality control processes, with all extreme values being checked and validated. It is well known that wind and snow cause errors in rain gauge

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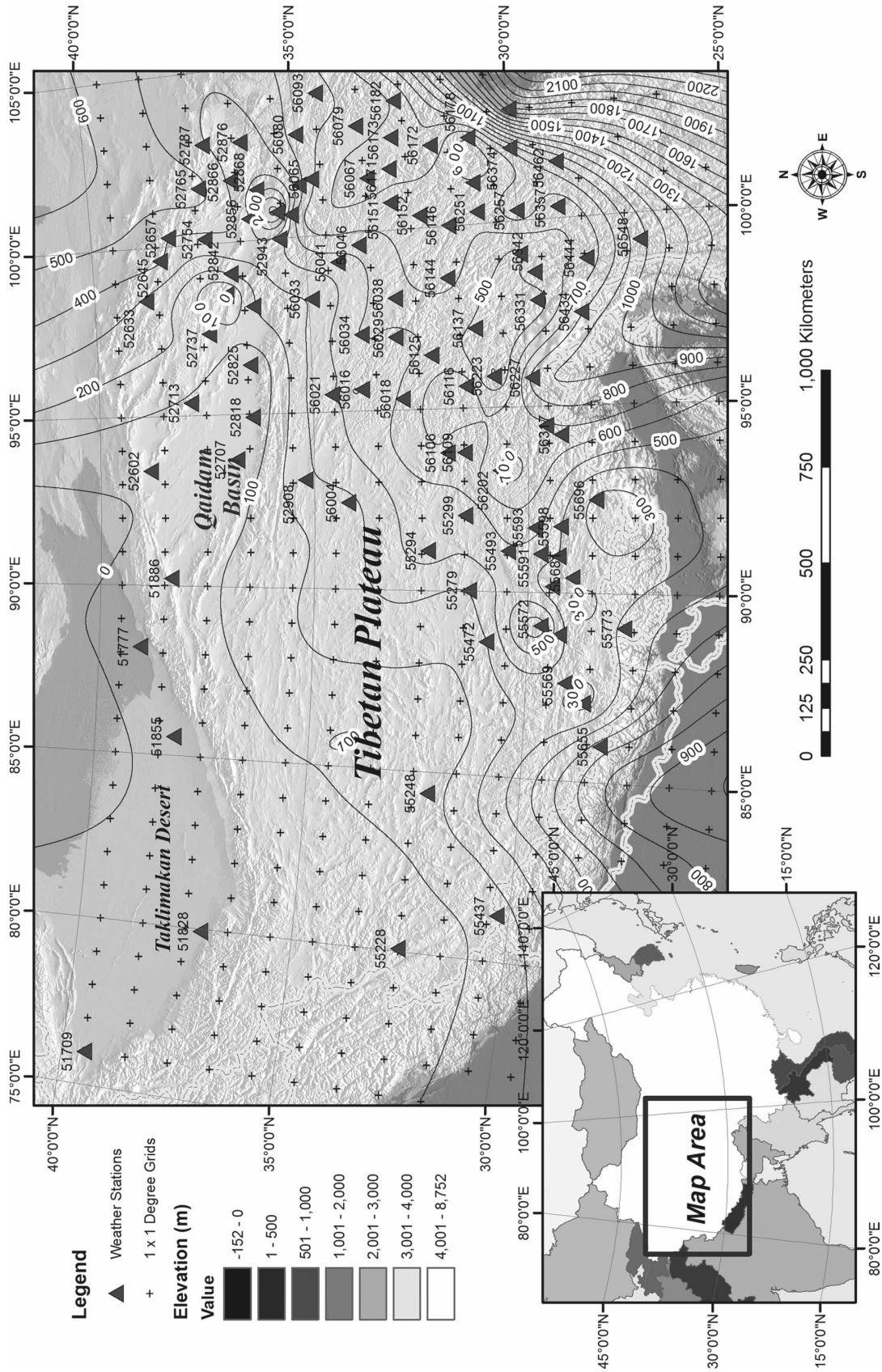


FIG. 1. Tibetan Plateau and the 94 weather stations used in this study. The stations are labeled with the standard station identification (ID) codes from the China Meteorological Administration. The crosses represent the central location of the 1° × 1° grids. The thick black lines are the mean annual isohyetals based on data during 1951–2004.

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data (Legates and Willmott 1990), especially when snow makes up a large portion of the precipitation. By avoiding the winter months (November to March next year), we hope this issue has been somewhat alleviated.

In this study, GIS was used to assemble spatial datasets, perform interpolation to generate surfaces of rainfall estimates, and to carry out topographic analysis and extract various information from digital elevation models (DEMs). The original SSM/I and TRMM 3B42 V5 satellite rainfall estimate data can be considered as the summary precipitation within the 1° × 1° box. Interpolation will eliminate the abrupt change along the border of the 1° × 1° boxes and generate a continuous surface for each month during the study period. Since precipitation over the Tibetan Plateau has a strong spatial trend (Fig. 1), decreasing from southeast to northwest, universal kriging with a linear drift trend was adopted as the interpolation method (Cressie 1993).

Since the monitoring of precipitation by satellites and that at weather stations have different spatial and temporal characteristics, it is difficult to match the two fields with perfect certainty. In this study, we used a downscaling method to take the advantage of the gridded satellite datasets, which produces relatively consistent quality of interpolation across the study region. To match the interpolated satellite precipitation data with the gauge data at the 94 weather stations used in this study, a buffer of 50 km was generated for each station. The size of the buffers matches the spatial resolution of the satellite data so that the spatial gradients and other features in the precipitation field would remain similar to that of the original dataset. Then values of raster surfaces, such as interpolated satellite estimates and topographic variables obtained from DEM, were extracted using the zonal statistics and tabulate area functions of ArcGIS (ESRI, Redlands, California). In this study, we considered three sets of variables that may potentially improve satellite-based rainfall estimates: 1) location variables (latitude and longitude) to determine whether there are spatial trends in performance of the algorithms; 2) terrain variables based on local topographic characteristics such as elevation, slope gradient, and aspect; and 3) fetch variables based on distances and directions to prominent topographic features of certain relief (500 or 1000 m above a generalized surface).

Using a similar approach as defined in Yin et al. (2004), topographic variables were generated using the 30 arc s global DEMs (GTOPO30) (http://edcdaac.usgs.gov/gtopo30/gtopo30.asp), resampled to 1 km × 1 km resolution. Elevation, slope aspect, and slope gradient are commonly considered topographic variables in studies of orographic effect (Basist et al. 1994; Daly et al. 1994; Johnson and Hanson 1995; Brown and Comrie 2002), which can all be obtained using the local and neighborhood analysis functions of ArcGIS. Fetch, in this study, is a term used to describe the spatial extent of surfaces with similar characteristics, especially in the upwind or downwind side of a mountain range across the path of moisture-carrying air flows. It denotes a set of variables based on distances and directions to prominent topographic features. To identify such topographic features, we first smoothed the DEM data by a 101 × 101 moving window or kernel. The size of the kernel matches the overall spatial scale of this study (1° × 1°) and the resultant smoothed surface represents large topographic features (50 km or larger). Then the resultant surface was subtracted from the original DEM to produce local relief to represent micro- to mesoscale features and identify locations with local relief higher than 500 or 1000 m. We also used the flow accumulation function of ArcGIS to identify cells of zero flow accumulation. These cells are often located in the headwater region without any upslope water collection area, corresponding to mountain peaks and ridgelines. Therefore, locations with local relief greater than 500 or 1000 m in combination of zero flow accumulation would represent mountain peaks and ridgelines. Based on the locations of these topographic features, a set of direction and distance variables can be extracted for the 50-km buffers. Figure 2, as an example of topographic analysis, shows the direction at any location toward mountain peaks or ridgelines with relief of 1000 m or higher (the inset) and the proportions of various such directions inside the 50-km buffers. The fetch variables should reflect topographic features known to influence precipitation in this region, such as density of, distance, and direction to mountain peaks and ridgelines (Barros et al. 2004, 2006).

Table 1 summarizes the variables considered in this study. Correlation and regression analyses were used to determine which location and topographic variables can make significant contributions to improve the satellite rainfall estimates. When sample size was relatively small and/or when there was any uncertainty regarding the assumption of linear relationships, the nonparametric Spearman's rank order correlation was used instead of the Pearson's product moment correlation.

Some of the topographic variables as defined in Table 1 are related to each other. For example, for an area with predominant southeastern slope aspect, there is a good chance that a high mountain peak or ridgeline is located in the northwestern direction close by. We employed principal component analysis (PCA) to reduce the redundancy in the topographic dataset (Johnston

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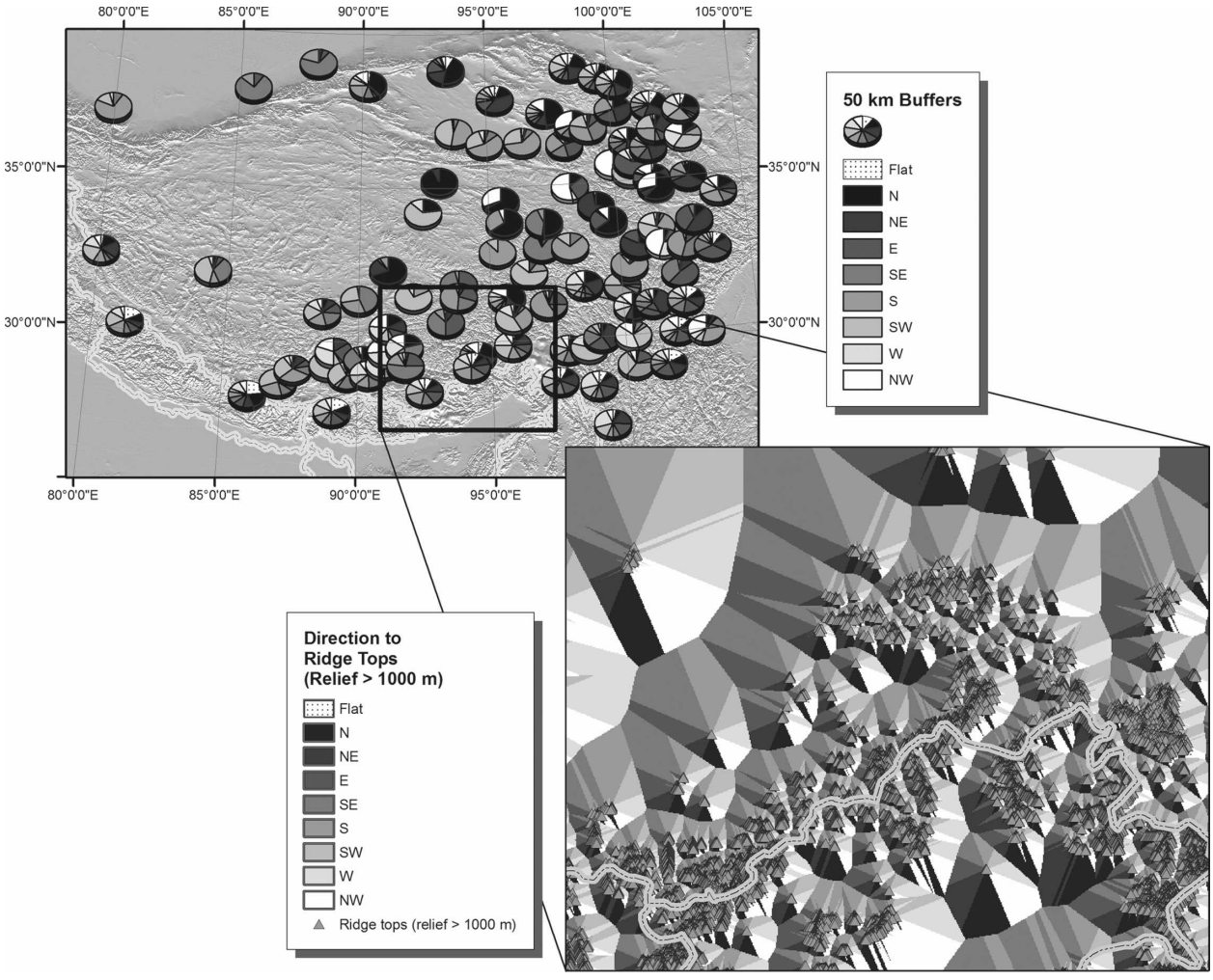


FIG. 2. An example of fetch variables obtained through topographic analysis using GIS: directions from any location to mountains or ridgelines with a relief of 1000 m or higher (the inset) and the proportions of such directions inside the 50-km buffers.

1978). For each topographic variable of terrain or fetch type, there are 94 observations corresponding to the number of weather stations. In PCA, the effect of similar variables will be combined in a new dataset with fewer variables known as principal components, or PCs. The first PC represents most variance in the original dataset. The second PC represents most variance in the dataset after the effect of the first PC is removed, and is orthogonal to the first PC. The process is continued until the subsequent PCs are not significant. In this study, the resultant PCs are rotated using the varimax method (Kaiser 1958), the most popular rotation method after which the rotated PCs (RPCs) are still orthogonal to each other, but with a "simple structure" by maximization of the number of RPCs with near-zero loadings (Richman 1986; White et al. 1991). As a result, each RPC would represent a more distinct group of original variables, as indicated by high loading values or the correlation coefficients with the original variables. In PCA, it is often subjective to determine how many PCs will be retained for further analysis. A commonly used method is the scree plot in which the turning point in the curve of the eigenvalue versus number of PCs retained is used to determine the cutoff point (Cattell 1966). Another method is to use the average eigenvalue of 1.0, or the Kaiser criterion, as the cutoff point (Kaiser 1960). In this study, we used a trial-and-error approach by performing a series of PCA procedures for different cutoff numbers, beginning with the number as suggested by the scree plot and the Kaiser criterion. After the rotation, we examined the matrix of the loading values, in which each RPC should represent at least one original variable. If there were RPCs that did not represent any original variable, the new dataset was considered "overfactored" and the number of RPCs of the new dataset was reduced.

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TABLE 1. Terrain and fetch variable definitions and descriptions. All variables are obtained using terrain analysis procedures in ArcGIS based on the 30 arc s DEM data (GTOPO30).

(a) Terrain variables—derived using the digital elevation models

Variable name	Description
Minelev	Minimum elev inside 50-km buffers
Maxelev	Maximum elev inside 50-km buffers
Rangelev	Range of elev values inside 50-km buffers
Meanelev	Mean elev inside 50-km buffers
Stdelev	Std dev of elev inside 50-km buffers
Meanslp	Mean slope angle inside 50-km buffers
Meanhshd	Mean lighting condition inside 50-km buffers, as represented by relative solar radiation with solar azimuth at 180° (south) and alt of 55°
Sumelev	Sum of all elev values inside 50-km buffers
Medelev	Median elev inside 50-km buffers
min_*h*	Minimum relative relief inside 50-km buffers, based on a 100-km search radius
max_*h*	Maximum relative relief inside 50-km buffers, based on a 100-km search radius
mean_*h*	Mean relative relief inside 50-km buffers, based on a 100-km search radius
std_*h*	Std dev of relative relief inside 50-km buffers
flt_asp	Proportion of flat terrain inside 50-km buffers, where slope aspect is coded as 0
n1_asp	Proportion of area with north-facing slopes inside 50-km buffers
ne2_asp	Proportion of area with northeast-facing slopes inside 50-km buffers
e3_asp	Proportion of area with east-facing slopes inside 50-km buffers
se4_asp	Proportion of area with southeast-facing slopes inside 50-km buffers
s5_asp	Proportion of area with south-facing slopes inside 50-km buffers
sw6_asp	Proportion of area with southwest-facing slopes inside 50-km buffers
w7_asp	Proportion of area with west-facing slopes inside 50-km buffers
nw8_asp	Proportion of area with northwest-facing slopes inside 50-km buffers

(b) Fetch variables—based on ridge tops with relative relief greater than 500 or 1000 m

Variable name	Description
flt_dir1	Proportion of the area inside 50-km buffers of no clear direction to ridge tops higher than 1000 m
n1_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the north
ne2_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the northeast
e3_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the east
se4_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the southeast
s5_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the south
sw6_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the southwest
w7_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the west
nw8_dir1	Proportion of the area where the ridge tops higher than 1000 m are in the northwest
flt_dir5	Proportion of the area inside 50-km buffers of no clear direction to ridge tops higher than 500 m
n1_dir5	Proportion of the area where the ridge tops higher than 500 m are in the north
ne2_dir5	Proportion of the area where the ridge tops higher than 500 m are in the northeast
e3_dir5	Proportion of the area where the ridge tops higher than 500 m are in the east
se4_dir5	Proportion of the area where the ridge tops higher than 500 m are in the southeast
s5_dir5	Proportion of the area where the ridge tops higher than 500 m are in the south
sw6_dir5	Proportion of the area where the ridge tops higher than 500 m are in the southwest
w7_dir5	Proportion of the area where the ridge tops higher than 500 m are in the west
nw8_dir5	Proportion of the area where the ridge tops higher than 500 m are in the northwest
mindis1	Minimum distance inside 50-km buffers to ridge tops higher than 1000 m
maxdis1	Maximum distance inside 50-km buffers to ridge tops higher than 1000 m
randis1	Range of distance inside 50-km buffers to ridge tops higher than 1000 m
meandis1	Mean distance inside 50-km buffers to ridge tops higher than 1000 m
stddis1	Standard deviation of distance inside 50-km buffers to ridge tops higher than 1000 m
mindis5	Minimum distance inside 50-km buffers to ridge tops higher than 500 m
maxdis5	Maximum distance inside 50-km buffers to ridge tops higher than 500 m
randis5	Range of distance inside 50-km buffers to ridge tops higher than 500 m
meandis5	Mean distance inside 50-km buffers to ridge tops higher than 500 m
stddis5	Std dev of distance inside 50-km buffers to ridge tops higher than 500 m

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TABLE 2. Descriptive statistics of the station precipitation, SSM/I and TRMM 3B42 V5 rainfall estimates, and absolute biases for April through October of the study period (mm). Here $N = 470$ for all variables.

Variable	Descriptive statistics	Month						
		Apr	May	Jun	Jul	Aug	Sep	Oct
Station P	Mean	24.6	49.6	84.4	100.2	104.7	63.7	27.8
	Std dev	30.8	39.4	56.7	65.5	73.9	41.1	31.0
	Min	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	258.8	208.2	275.6	374.2	543.8	258.2	237.6
SSM/I	Mean	4.4	22.4	69.6	81.2	68.5	45.6	2.7
	Std dev	19.0	45.2	74.4	63.5	58.9	76.2	8.6
	Min	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	298.9	352.4	390.3	292.8	397.5	613.3	93.4
TRMM	Mean	99.7	84.4	93.5	125.6	109.9	60.6	40.6
3B42 V5	Std dev	70.9	49.9	65.8	93.4	78.6	41.3	31.7
	Min	2.6	0.0	3.7	0.0	0.0	0.0	0.0
	Max	450.2	320.6	306.6	399.6	370.0	206.2	217.6
Bias of	Mean	−20.1	−27.2	−14.8	−19.0	−36.2	−18.1	−25.1
SSM/I	Std dev	32.5	46.5	61.8	53.3	62.6	70.5	30.7
	Min	−258.1	−184.0	−207.8	−279.9	−449.4	−161.7	−237.6
	Max	168.9	246.4	346.0	135.1	162.6	524.2	39.1
Bias of	Mean	75.2	34.7	9.1	25.4	5.2	−3.1	12.8
TRMM	Std dev	71.7	47.5	56.5	70.4	54.0	36.9	33.7
3B42 V5	Min	−132.8	−90.3	−124.8	−163.9	−372.9	−173.1	−141.3
	Max	384.8	232.9	215.2	291.4	203.0	113.9	190.1

3. Results and discussion

a. Performance of the satellite rainfall estimates

Table 2 lists descriptive statistics of monthly station precipitation, SSM/I and TRMM 3B42 V5 estimates, and absolute biases averaged for April through October for the study period, with data from all stations pooled together. Although the satellite estimates represented the long-term (5 yr) average of the seasonal cycle fairly well, SSM/I consistently underestimated and TRMM 3B42 V5 overestimated the station rainfall amount. For the summer months, the satellite estimates on average performed better than during the spring and fall months.

Figure 3 shows the spatial patterns of the mean relative biases of SSM/I and TRMM 3B42 V5 rainfall estimates as ratios to the station observations, averaged for all months during 1998–2002. In general, SSM/I underestimated the station precipitation for a large central portion of the study region, while TRMM 3B42 V5 overestimated in areas north and south of the region. The area where the satellite estimates performed best is in the east of the study region. By further examining the relative biases for individual months, we noticed that the spatial patterns were changing from month to month although certain characteristics remained persistently (Fig. 4). In April, May, and October, SSM/I grossly underestimated station precipitation across the entire region, while TRMM 3B42 V5 mainly overestimated station precipitation. In June and July, the SSM/I improved significantly and only underestimated station values in a smaller interior portion of the plateau. Likewise, TRMM 3B42 V5 overestimated station values only along the southern margin of the plateau. In August, the TRMM 3B42 V5 provided the best performance, but still overestimated along the southern margin and at a few stations to the north in the Qaidam Basin area. SSM/I, on the other hand, had a pattern of underestimating along a belt in the interior running northeast–southwest. The September's patterns were similar to those in June.

It seems that TRMM 3B42 V5 tended to overestimate precipitation persistently along the southern margin of the study region. Barros et al. (2006) found that the TRMM PR had difficulties in detecting precipitation at high elevations, especially in the rain shadows. However, this effect may have been somehow overcompensated by the TMI or IR component in the 3B42 V5 dataset. Additionally, low precipitation amount in the rain shadow of the Himalayas would make the error of margin more prominent corresponding to the same sensor sensitivities. In the mean time, SSM/I consistently underestimated station values in the interior part of the study region. The elevation of the plateau surface is over 4000 m MSL. It is possible that the presence of permafrost and desertlike tundra surface caused the NESDIS algorithm to underestimate (Ferraro et al. 1998). The plot of the mean biases of all stations by

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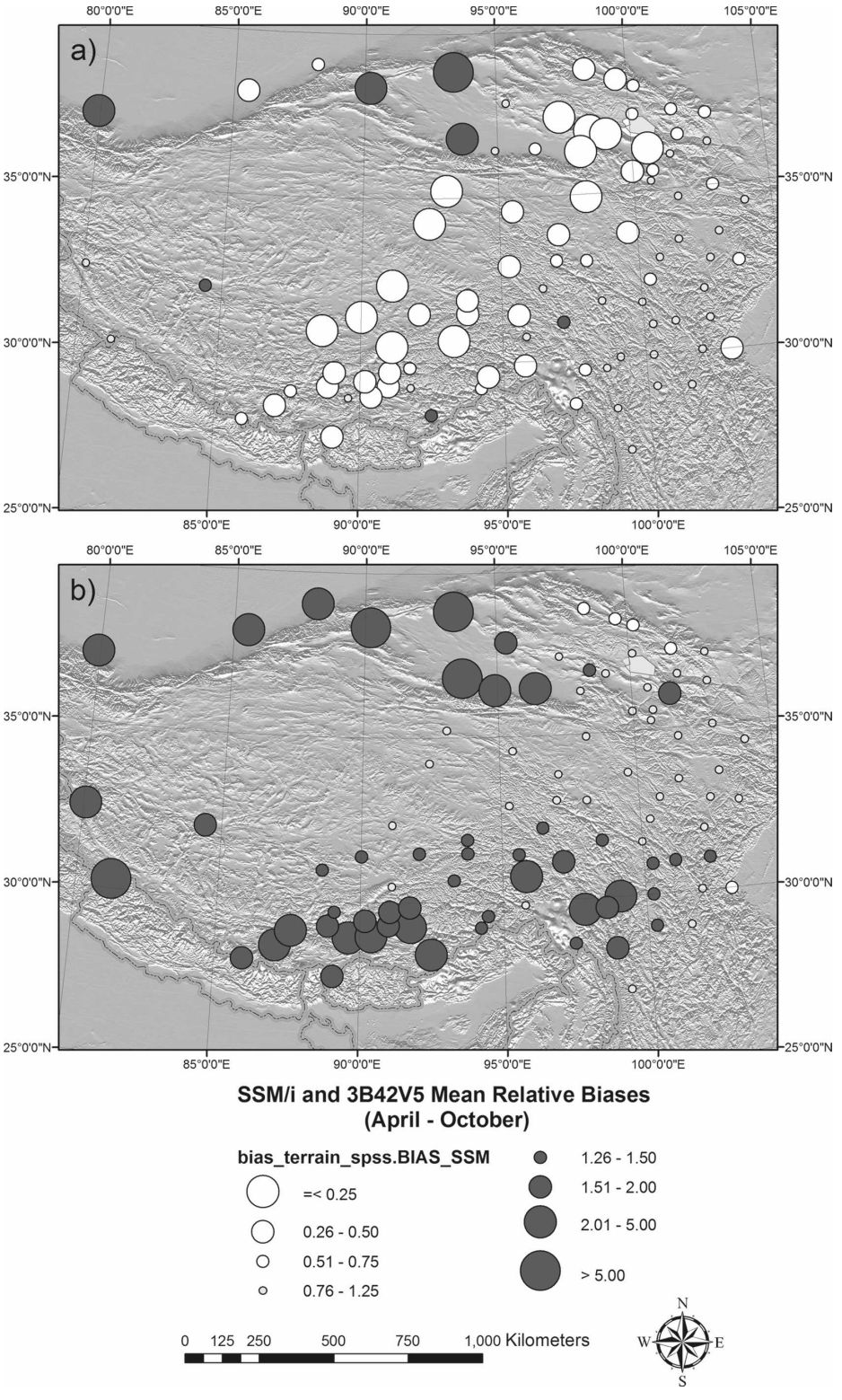


FIG. 3. Mean relative biases of (a) SSM/I and (b) TRMM 3B42 V5 rainfall estimates as the ratios to station observations, with all months from April to October during 1998–2002 pooled together.

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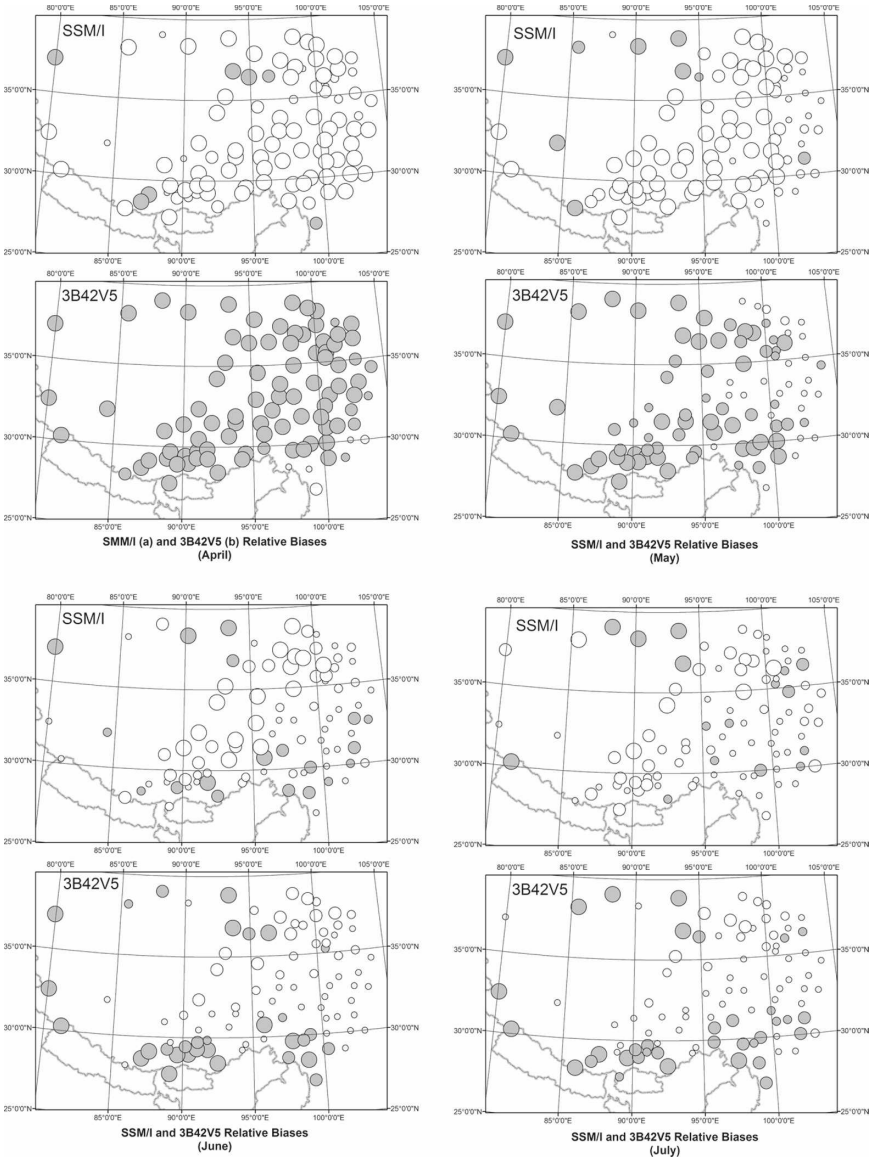


FIG. 4. Spatial patterns of the relative biases of SSM/I and TRMM 3B42 V5 rainfall estimates by individual months. Some stations were missing because relative biases could not be calculated for stations with observed zero precipitation values.

months (Fig. 5) shows that TRMM in April and October had the largest relative biases. Our results suggest that the satellite data for April and October should be used with great caution. A closer examination found that the very high mean relative biases of these months were caused by gross overestimation at a few stations in combination with very low observed rainfall amount. For this reason, the descriptive statistics of the absolute biases presented in Table 2 offered a more realistic assessment for the spring and fall months. The performance of TRMM 3B42 V5 during the summer months was much better compared to that in the spring months, with August rendering the best estimates. In contrast, the relative biases of SSM/I were much more stable for April through October, but mostly below 1.0 or underestimating except for May and June. It should be noted that when all monthly values were pooled together as in the case of Table 2, the results were slightly different from those when the estimates were first averaged for the stations over the 5-yr period, although the general patterns remained the same.

Another measure of performance is to examine how the satellite estimates covaried with the observed precipitation values. We calculated the Pearson's product

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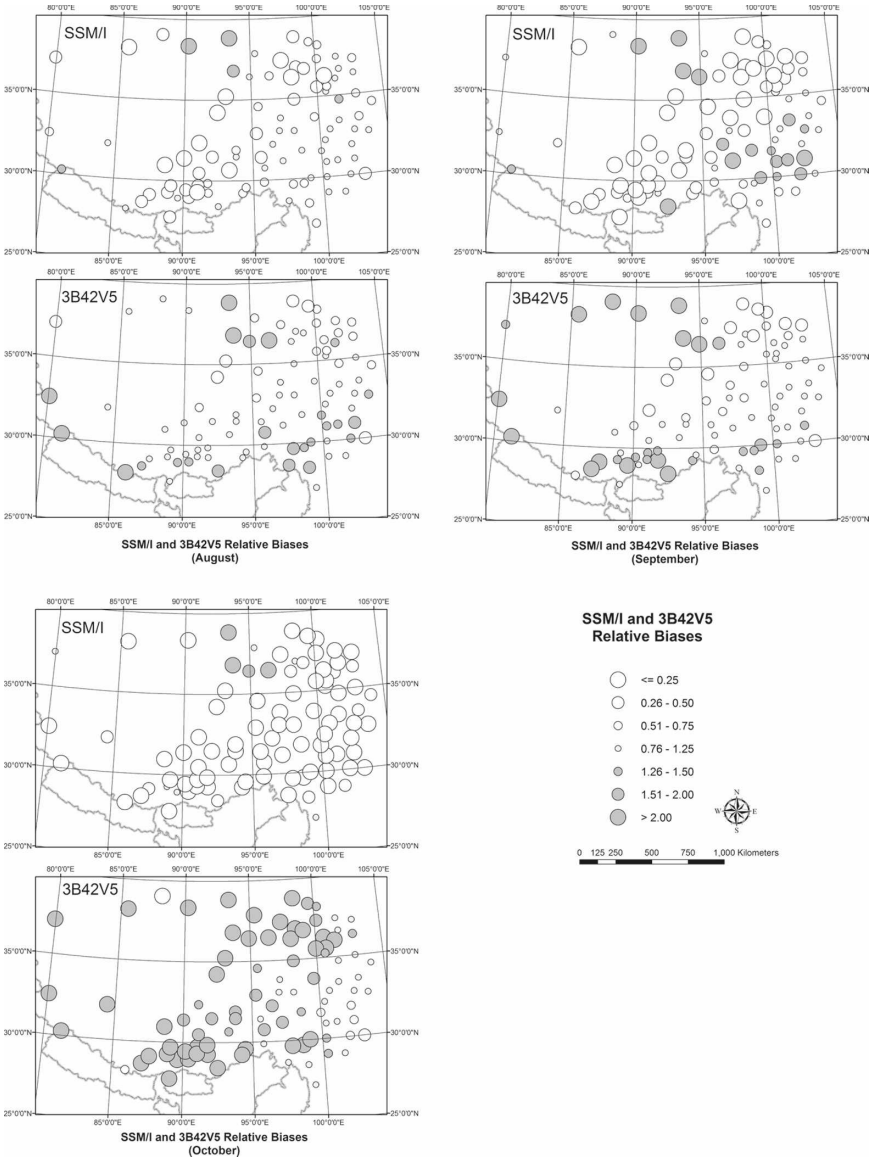


FIG. 4. (*Continued*)

moment correlation coefficients between the observed and satellite-estimated values by individual months with all station data pooled together. The best performance was again seen during the summer months when most rainfall occurred. Although still statistically significant, the satellite estimates in April and October correlated poorly with the observed values (Fig. 6a). It is also interesting to see the "decoupling" between the SSM/I and TRMM 3B42 V5 data in spring and fall months (Fig. 6b). Figure 7 shows the Spearman's rank correlation coefficients between station-observed monthly precipitation to SSM/I and TRMM 3B42 V5 estimates for April through October during 1998–2002 ($N = 35$). This should indicate how satellite estimates represent seasonal, as well as interannual, variation patterns in the study region. The SSM/I estimates showed statistically significant correlations at most stations. However, the performance of TRMM 3B42 V5 estimates appeared to be extremely regionalized, with strong correlations to the observed precipitation values in eastern and southern parts of the region, but with weak correlations in the interior of the plateau.

b. Principal component analysis of terrain and fetch variables

In topographic analysis, we extracted a total of 50 terrain and fetch variables that potentially have some effects on rainfall distribution across the study area

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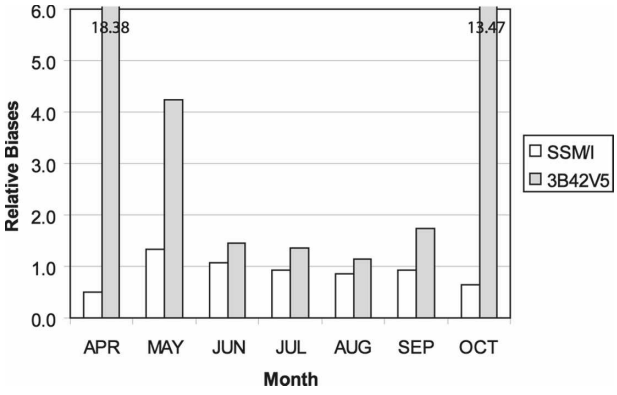


FIG. 5. Relative biases of the satellite estimates averaged by month during 1998–2002. The overestimation by TRMM 3B42 V5 in April and October were over 18 times and 13 times, respectively, but truncated in the graph.

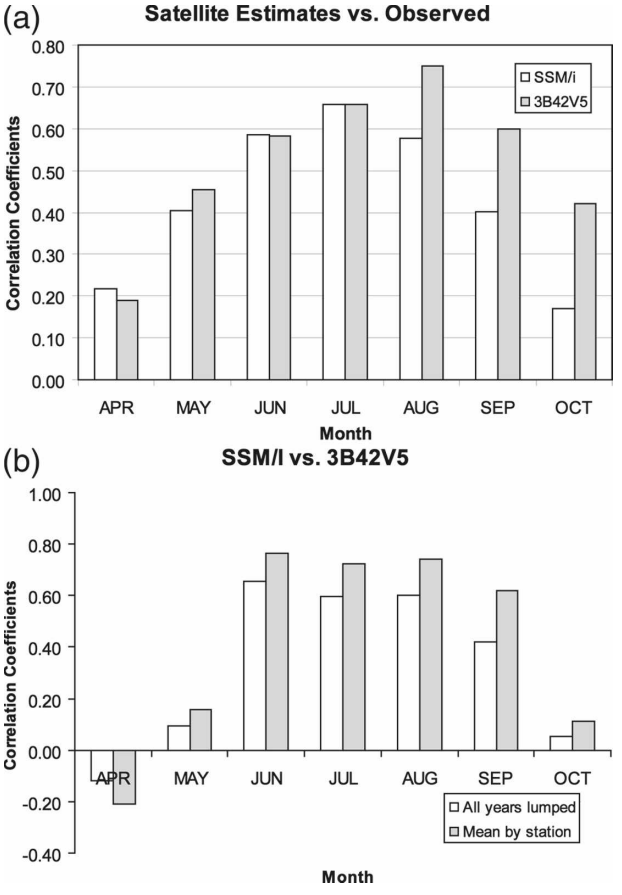


FIG. 6. (a) Pearson's product moment correlation coefficients between station rainfall and satellite estimates by months and (b) the correlations between SSM/I and TRMM 3B42 V5 estimates by months, with all station data pooled together ($N = 470$).

(Table 1). To select those variables that can make statistically significant contributions to predicting the spatial pattern of precipitation, we conducted correlation analysis using station precipitation data for April through October against the topographic variables. Forty-eight terrain and fetch variables had statistically significant correlation coefficients (at the 0.01 level) with the station precipitation for at least 1 month when all 5 yr of data were pooled together ($N = 470$). The two variables that did not have any significant correlation coefficients were nw8_asp and se4_dir1, which were excluded from the following PCA procedure. April had the most terrain and fetch variables that were correlated with station precipitation, while October had the least (Fig. 8).

In PCA, we determined that 16 rotated PCs (RPCs) rendered the best result, which explained more than 92% of variance of the original dataset and each RPC represented at least one original terrain or fetch variable, as indicated by the highest loading value (Table 3). Scores of the RPCs were calculated and used in regression analysis to correct satellite estimates (appendix B). Also listed in Table 3 are the terrain and/or fetch variables represented by each RPC, which helps identify topographic features important in improving satellite estimates.

c. *Correction of satellite estimates based on regression analysis*

We ran regression analysis first using only the SSM/I and TRMM 3B42 V5 estimates as the independent variables, then we added the location variables (latitude and longitude) to the models, and finally we added the RPCs representing the terrain and fetch variables. The SSM/I and TRMM 3B42 V5 estimates were forced into the models in each run, while all other variables were selected by the stepwise procedure. Only variables that were statistically significant at the 0.05 level were allowed to enter and remain in the models.

Table 4 presents the result of stepwise regression on station precipitation. With each step, the R^2 values or explained variances improved. For stepwise regression using location variables, both location variables entered all models except the one for August, in which only longitude was included. On average, the adjusted R^2 values increased from a mean of 0.259 when using either SSM/I or TRMM 3B42 V5 alone to 0.363 when both SSM/I and TRMM 3B42 V5 were used, to 0.468 when location variables were added, and to 0.546 when the terrain and fetch RPCs were added to the models. As a comparison, the reported correlations between satellite estimates (based on microwave, GPI, and outgoing longwave radiation for both over land and over ocean) and observation precipitation in earlier intercomparison studies would be equivalent to R^2 values of

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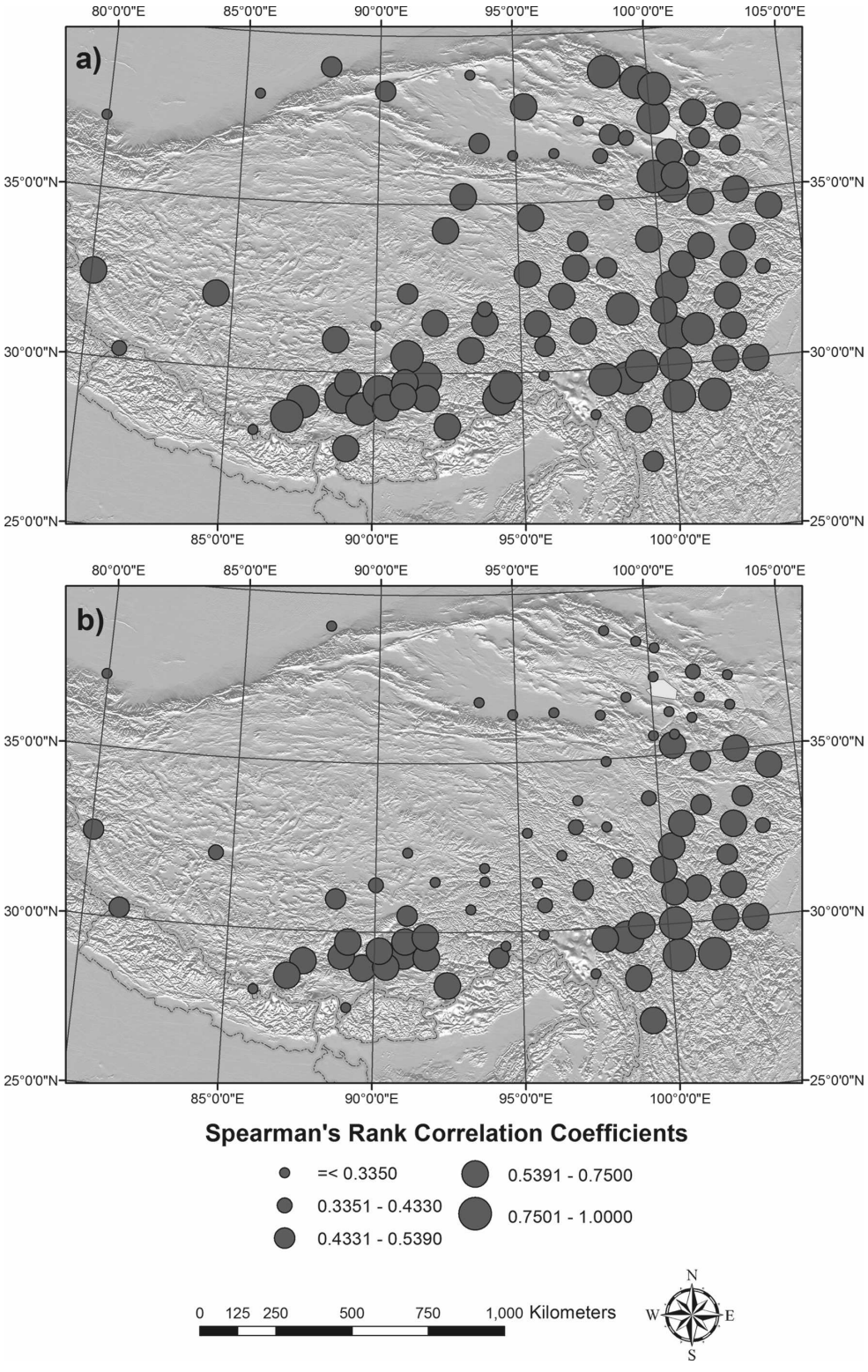


FIG. 7. Spearman's rank correlations of the (a) SSM/I and (b) TRMM 3B42 V5 estimates to the station observations of April through October during 1998–2002 ($N = 35$). The legend indicates the statistical significance of the correlation coefficients: 0.335 at the 0.05 level, 0.433 at the 0.01 level, and 0.539 at the 0.001 level (Ramsey 1989).

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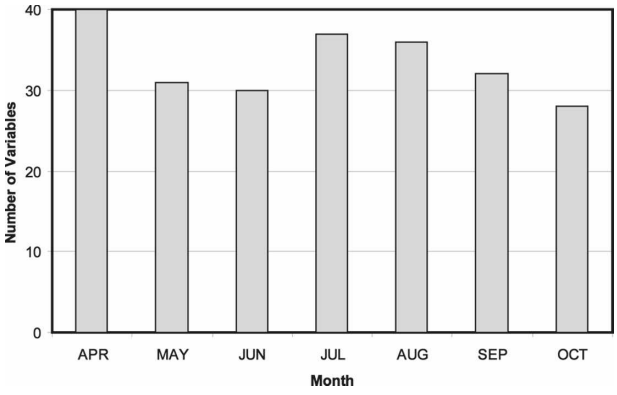


FIG. 8. Number of terrain and fetch variables that have significant correlations with precipitation in individual months of April through October.

0.3–0.6 in most cases (Arkin and Xie 1994; Xie and Arkin 1995, 1998; Conner and Petty 1998; Adler et al. 2001; Sealy et al. 2003). The most frequent RPCs included in the final models in stepwise regression are RPC5 (6 times), RPC1 and RPC2 (5 times each), RPC8 (4 times), and RPC3, RPC4, RPC6, and RPC9 (2 times each). Among these RPCs, RPC1 represents mostly variability of elevation or surface roughness; RPC2 is a measure of elevation overall associated with the forced ascent of topographic forcing; RPC5 represents slope aspect, especially the southeast-facing slopes toward the incoming moisture-laden air flows; and RPC8 represents the distance to peaks and ridgelines of 1000 m or higher as a measure of density of mountain peaks and ridgelines. Table 5 contains the regression models to illustrate how these topographic RPCs can be used as correction factors with different monthly weights (the regression coefficients) to improve SSM/I and TRMM precipitation estimates.

We plotted the monthly values of the SSM/I and TRMM 3B42 V5 estimates, and the model-predicted values against the station observed values in Fig. 9. The improvement over the original satellite estimates is quite obvious. The terrain-corrected satellite estimates explained over 71.0% of the variance in the observed precipitation, as compared to 38.2% by SSM/I and 34.1% by TRMM 3B42 V5 alone. We mapped the mean residuals of the regression models for each station (Fig. 10a), and there did not appear to be any significant spatial patterns to suggest systematic errors in the regression models. We also mapped the ratios of the mean model-predicted values to the observed values averaged by stations (Fig. 10b), which is a measure similar to the mean relative biases as presented in Fig. 3. We used the same set of symbols/legends for the purpose of comparison between the two figures. Although there are locations in the northern (the Qaidam Basin) and western parts of the region where large biases are still seen, significant improvements have been made for most of the study region.

d. A comparison with the TRMM 3B43 version 6 data

While the focus of this study is the 3B42 version 5 dataset, we also performed a comparison with the 3B43 version 6 data (http://daac.gsfc.nasa.gov/precipitation/

TABLE 3. Results of rotated principal component analysis with eigenvalues for each RPC and the terrain and fetch variables it represents. The terrain and fetch variables are listed under each RPC from the most negative to the most positive loading values. The number of variables used in PCA is 48.

	RPC1	RPC2	RPC3	RPC4	RPC5	RPC6	RPC7	RPC8
Eigenvalues	11.320	5.337	3.290	3.078	3.047	2.909	2.365	1.949
Variables Represented	randis1 stddis1 maxdis1 min_*h* meanhshd meanslp flt_dir1 flt_dir5 std_*h* max_*h* stdelev rangelev	minelev maxelev sumelev medelev meanelev	s5_dir5 e3_dir500 w7_asp ne2_dir5 sw6_asp	n1_dir5 s5_asp	n1_asp e3_asp nw8_dir5 se4_asp	meandis5 maxdis5 stddis5 randis5	se4_dir5 ne2_asp sw6_dir5	meandis1 mindis1
	RPC9	RPC10	RPC11	RPC12	RPC13	RPC14	RPC15	RPC16
Eigenvalues	1.721	1.499	1.442	1.380	1.336	1.327	1.215	1.110
Variables Represented	nw8_dir1 w7_dir1	mindis5	e3_dir1	s5_dir1 n1_dir1	sw6_dir1	w7_dir5 ne2_dir1	flt_asp	mean_*h*

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TABLE 4. Results of regression analysis using the scores of the RPCs. All models are based on 5 yr of data pooled together for individual months ($N = 470$).

(a) R^2 of regression models: station precipitation vs either SSM/I or TRMM 3B42 V5							
	Apr	May	Jun	Jul	Aug	Sep	Oct
SSM/I	0.048	0.163	0.342	0.434	0.332	0.161	0.029
TRMM 3B42 V5	0.036	0.207	0.340	0.434	0.564	0.360	0.177
(b) R^2 of regression models: station precipitation vs both SSM/I and TRMM 3B42 V5							
	Apr	May	Jun	Jul	Aug	Sep	Oct
R^2	0.095	0.337	0.412	0.543	0.588	0.386	0.199
Adjusted R^2	0.091	0.335	0.409	0.541	0.586	0.384	0.195
(c) R^2 of regression models: station precipitation vs satellite estimates (forced) and location variables (stepwise)							
	Apr	May	Jun	Jul	Aug	Sep	Oct
R^2	0.315	0.480	0.496	0.573	0.598	0.500	0.348
Adjusted R^2	0.309	0.475	0.492	0.569	0.595	0.496	0.342
Variables entered	long1	long1	long1	long1	long1	long1	*long1*
	lat1	lat1	lat1	lat1		lat1	lat1
(d) R^2 of regression models: station precipitation vs satellite estimates (forced), location, and topographic RPCs (stepwise)							
	Apr	May	Jun	Jul	Aug	Sep	Oct*
R^2	0.495	0.508	0.569	0.637	0.686	0.559	0.426
Adjusted R^2	0.486	0.502	0.562	0.63	0.678	0.552	0.414
Location	long1	long1	long1	long1	long1	long1	long1
Variables	lat1	lat1		lat1	lat1	lat1	lat1
RPCs entered	1, 2, 5, 6	5, 8	1, 2, 5, 8	1, 2, 3, 4, 9	1, 2, 3, 4,, 5, 10, 15	5, 8, 13	1, 2, 5, 6, 8, 9

* SSM/I is not statistically significant at the 0.05 level.

TRMM_README/TRMM_3B43_readme.shtml). This dataset is based on the 3B42 V6, which incorporates TRMM, SSM/I, and AMSU data, and then combined with the gauge analysis data of Climate Anomaly Monitoring System (CAMS) or GPCP (Huffman et al. 2007). Theoretically it should offer better performance than the SSM/I or TRMM 3B42 estimates. Using the same methods, the 0.25° × 0.25° 3B43 V6 data were interpolated and matched to the station 50-km buffer zones. When long-term mean relative biases were plotted across the study area (Fig. 11), the improvements over the 3B42 V5 are obvious as compared with Fig. 3. However, the spatial pattern of overestimation in the southern and the northeastern parts of the plateau remained. Regression analysis revealed that topographic and location variables can further improve the 3B43

TABLE 5. Regression models to predict station precipitation using satellite estimates, geographic location, and RPCs of terrain and fetch variables. All independent variables in the models are statistically significant at the 0.05 level.

Month	Regression models
Apr	−43.245 + 0.130 SSM/I + 0.079 TRMM* + 1.863 long1 −3.623 lat1 + 7.971 RPC1 −9.950 RPC2 + 3.549 RPC5 + 3.280 RPC6
May	−174.218 + 0.199 SSM/I + 0.207 TRMM + 2.739 long1 −1.825 lat1 + 4.474 RPC5 + 5.203 RPC8
Jun	−274.224 + 0.173 SSM/I + 0.279 TRMM + 3.352 long1 −3.979 RPC1 + 8.702 RPC2 + 8.086 RPC5 + 11.378 RPC8
Jul	171.035 + 0.402 SSM/I + 0.140 TRMM + 1.944 long1 −9.347 lat1 −14.558 RPC1 −10.367 RPC2 −6.387 RPC3 + 9.239 RPC4 + 5.295 RPC9
Aug	38.967 + 0.177 SSM/I + 0.591 TRMM + 1.743 long1 −5.419lat1 −17.099 RPC1 −12.343 RPC2 −9.125 RPC3 + 8.238 RPC4 + 5.630 RPC5−5.065 RPC10 + 4.261 RPC15
Sep	−123.135 + 0.050 SSM/I + 0.289 TRMM + 2.560 long1 −2.372 lat1 + 8.026 RPC5 + 5.666 RPC8 −3.076 RPC13
Oct	−118.565 + 0.128 SSM/I** + 0.252 TRMM + 1.884 long1 −1.363 lat1 + 3.144 RPC1 −4.079 RPC4 + 6.337 RPC5 + 3.637 RPC6 + 2.248 RPC8 −2.929 RPC9

* TRMM is the TRMM 3B42 V5 data.
** SSM/I is not statistically significant at the 0.05 level.

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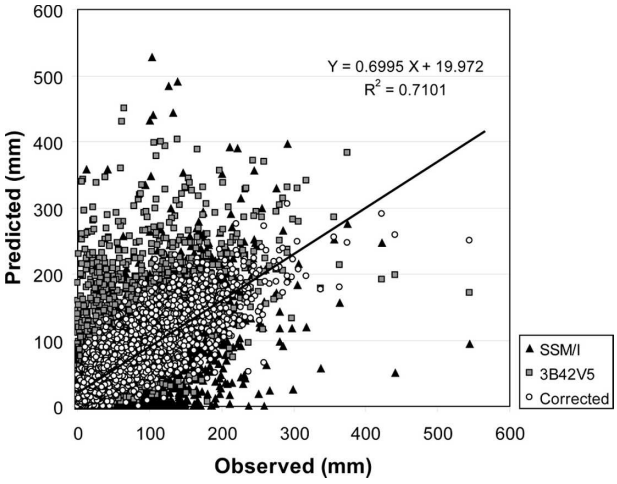


FIG. 9. The SSM/I, TRMM 3B42 V5, and model estimates plotted against the observed monthly precipitation values at all stations, 1998–2002.

data for all the months considered (Table 6). When compared with Table 4, it is clear that some rotated PCs consistently entered the regression models, such as RPC1, PRC5, RPC6 (distance to peaks and ridgelines of 500 m or higher and its variability), RPC8, and RPC9 (peaks and ridgelines of 1000 m or higher located to the west or northwest). When the original 3B43 V6 values and the corrected estimates were compared with the observed station precipitation, the R^2 values increased from 0.792 to 0.844 for all months and stations being pooled together, and the advantage of the terrain-corrected values was clearly demonstrated in Fig. 12. This result suggests that there is still the need for correcting even the latest version of satellite estimates using terrain and fetch variables generated by GIS.

4. Discussion

There have been numerous studies on the orographic effect of topographic features on precipitation. For example, summarizing previous works, Roe et al. (2003) described the mechanisms of orographic precipitation, including forced ascent on the windward slopes and adiabatic warming on the leeward slopes when mountain ranges lie across the prevailing moisture-laden wind directions, enhancement of synoptic-scale rainfall events by low-level cloud droplets corresponding to locations of mountains and valleys, cloud formation due to the combination of forced ascent and conditionally unstable atmosphere to cause an air parcel to rise above its level of free convection, and abruptly increased roughness along the paths of the airflow (Smith 1979; Bergeron 1960; Barros and Lettenmaier 1994). Orographic precipitation may have strong regional and seasonal characteristics. Barros et al. (2004) found a strong contrast between the Northern Indian plains and Tibetan Plateau in diurnal patterns of convective cloud clusters and disorganized short-lived convection development. Barros et al. (2004, 2006) also identified features of different spatial scales, from small- to mesoscale features associated with the succession of ridges and valleys, to synoptic features controlled by the overall topography of the Himalayas and major river valleys that cut through the mountains and connect the Indian subcontinent and Tibetan Plateau.

Following the results of a study in western Canada (Klimeš 1990), there should be a dramatic reduction of precipitation in the rain shadow north of the Himalayas, followed by fluctuation of high (low) precipitation on the southeast-facing (northwest-facing) slopes for mountain ranges across the Tibetan Plateau, and a general reducing trend from southeast to northwest. Because of the limitation of spatial and temporal resolution of the datasets, however, our analyses could not fully identify factors at the micro- to mesoscales and explain the variation patterns of the biases based on physical processes. However, our study revealed topographic variables, easily extracted from the DEMs, which contributed significantly in correction of the biases in satellite rainfall estimates. Four types of topographic variables entered the regression models most frequently: 1) those that reflect the variability of elevation or surface roughness, such as slope gradient, standard deviation, and range of elevation or relief as represented by RPC1; 2) those that reflect overall elevation associated with forced ascent as represented by RPC2; 3) those that reflect slope aspects facing toward the pathway of moisture-laden air flows as represented by RPC5; and 4) those that reflect the location and direction of prominent topographic features such as mountain peaks and ridgelines as represented by RPC6, RPC8, and RPC9.

5. Conclusions

In this study we evaluated the spatial and temporal patterns of the biases of monthly rainfall estimates based on SSM/I, TRMM 3B42 V5, and 3B43 V6 datasets over the Tibetan Plateau for the months of April through October during 1998–2002. The satellite estimates were downscaled by interpolation to match with observed monthly precipitation values at 94 stations across the study region. The high altitude and complex topographic conditions over the plateau imposed unique challenges to both datasets. In comparison, the TRMM 3B42 V5 product rendered consistently

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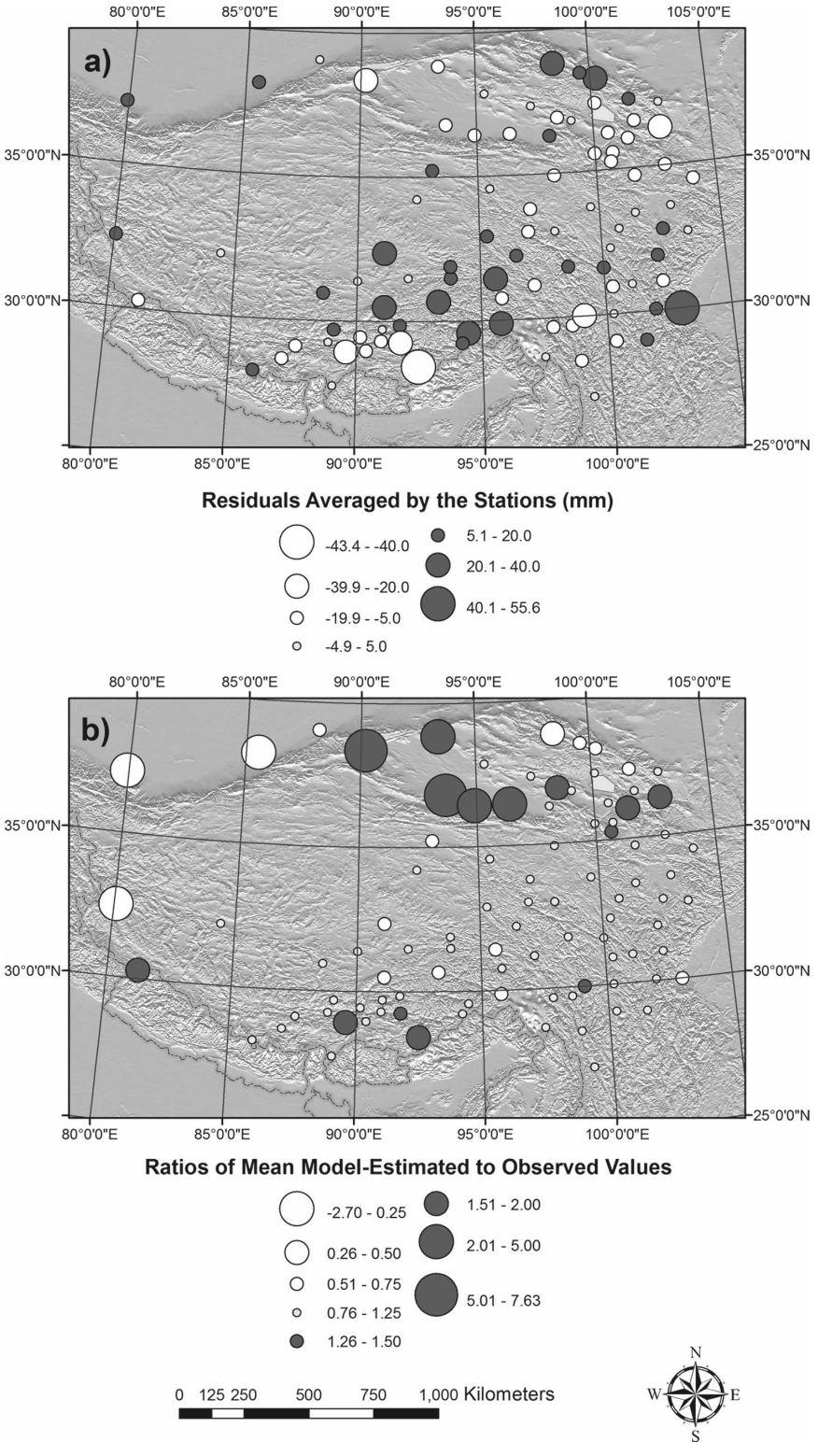


FIG. 10. (a) Spatial patterns of the residuals averaged by stations and (b) the ratios of the mean model-predicted values to mean station precipitation averaged for all months of April–October during 1998–2002.

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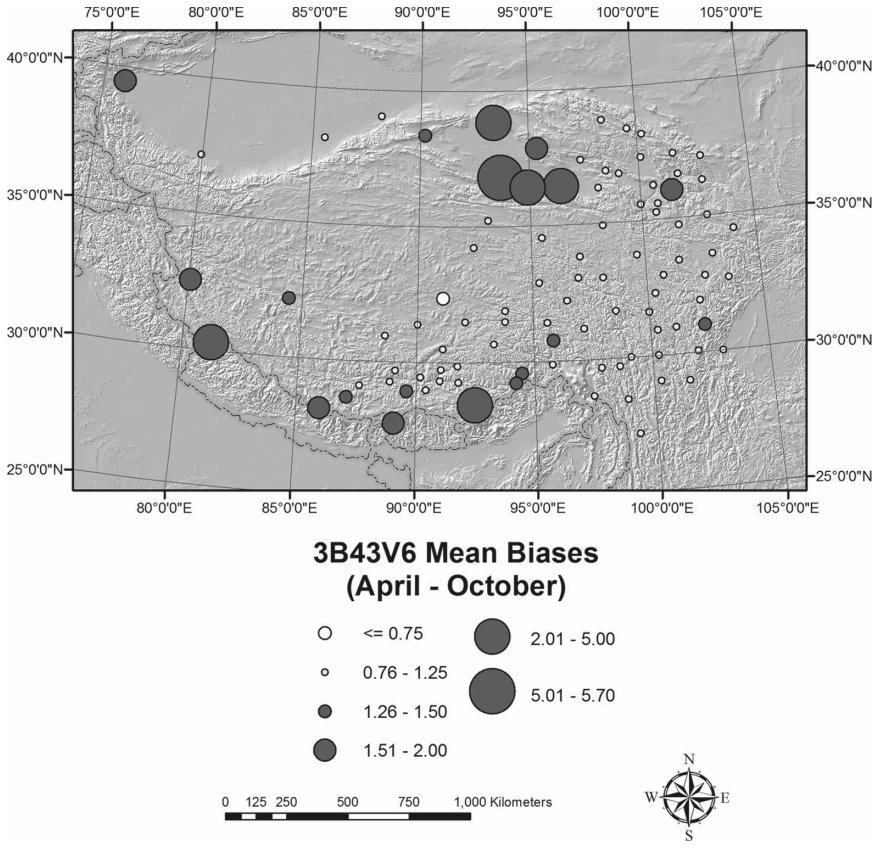


FIG. 11. Spatial pattern of the relative biases of the 3B43 V6 dataset across the Tibetan Plateau, with estimates as ratios to station precipitation with all months from April to October during 1998–2002 pooled together.

higher values than the SSM/I estimates, although these two datasets produced similar performance measured by the results of regression analysis against the observed values. The biases and performance of the TRMM 3B42 V5 data have distinct regional patterns, with stronger correlations to but also greater overestimation of the observed values in the southeast and south of the study region. The SSM/I underestimated the rainfall amount for most parts of the study region, especially along a belt running northeast–southwest at the center, but its overall performance is less variable spatially and temporarily than TRMM 3B42 V5. The spatial patterns of the biases changed from month to month, suggesting certain seasonal patterns

TABLE 6. Results of regression analysis for TRMM 3B43 V6 data by month.

(a) R^2 of regression models: station precipitation vs TRMM 3B43 V6							
	Apr	May	Jun	Jul	Aug	Sep	Oct
R^2	0.689	0.718	0.737	0.735	0.720	0.719	0.706
Adjusted R^2	0.688	0.718	0.736	0.734	0.720	0.718	0.705
(b) R^2 of regression models: station precipitation vs TRMM 3B43 V6, location, and topographic RPCs (stepwise)							
	Apr	May	Jun	Jul	Aug	Sep	Oct
R^2	0.712	0.764	0.801	0.812	0.793	0.771	0.718
Adjusted R^2	0.709	0.761	0.799	0.809	0.790	0.767	0.716
Location variables		long1 lat1	long1	lat1		long1 lat1	long1
RPCs entered	1, 2, 16	1, 5, 8	1, 8, 9, 11	1, 6, 9, 16	1, 2, 5, 6, 9, 16	1, 5, 8, 9	5, 6

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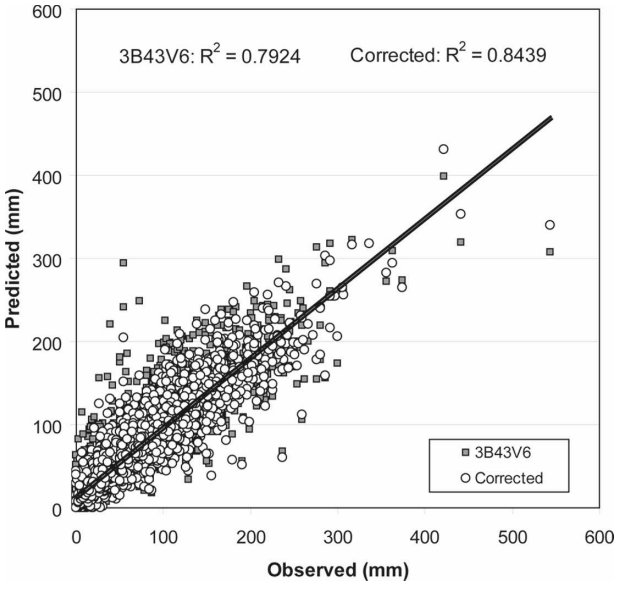


FIG. 12. The original and corrected 3B43 V6 data plotted against the observed monthly precipitation values with all months from April to October during 1998–2002. The trend line and equation are for the corrected values.

of the influencing factors. Simply by using both SSM/I and TRMM 3B42 V5 together in rainfall estimation, the overall performance can be significantly enhanced.

To identify variables that can be used as correction factors to improve satellite rainfall estimates over the Tibetan Plateau, we used GIS to perform topographic analysis and extracted 50 different terrain and fetch variables from the 1-km-resolution digital elevation models. Results of principal component analysis defined 16 rotated principal components, which were used together with satellite rainfall estimates and location variables in regression analysis against the observed station precipitation. The location variables, latitude and longitude of weather stations, indicated strong spatial trends in the performance of satellite rainfall estimates. By adding the rotated principal components representing the topographic variables to the regression models, the performance of the satellite rainfall estimates can be improved significantly. These topographic variables reflected surface roughness, height or relief of topographic features, slopes facing toward the source of moisture, and relative location or direction to prominent topographic features such as mountain peaks and ridgelines.

With the results from this study, it is feasible to use the rotated principal components of the terrain and fetch variables as the correction factors to improve satellite estimates. The weights (regression coefficients) of these correction factors, which changed over time, reflect the relative significance of the correction factors for different months. It should be pointed out that our approach in correcting satellite rainfall estimates is entirely empirical as limited by the spatial and temporal resolutions of the datasets examined in this analysis. The temporal scale (monthly data) of this study should somewhat compensate the effects of low frequency of revisit of the satellites and gaps of spatial coverage. However, such integration would limit our analysis to identify specific physical processes as the forcing mechanisms of the orographic effect. The results can be very well explained by general understanding of local variation patterns of climate and orographic effect of topographic features. For example, the variables that entered regression models most frequently represented the variability of elevation or surface roughness, density of peaks and ridgelines, slope aspects facing south or southeast, and peaks and ridgelines located to the west or northwest of a given location. All these have been directly or indirectly suggested as being important topographic factors of orographic precipitation (Barros et al. 2004, 2006). Even so, cautions must be exerted when using such empirical relationships in areas without ground observations. However, our ground stations covered a wide range of different topographic conditions across the entire Tibetan Plateau, which offered the confidence to use such relationships to improve satellite rainfall estimates in an operational fashion. There should be plenty of work still to be done in this region. One uncertainty resulted from the downscaling procedure that may not fully make the point measurements at the weather stations comparable to the satellite estimates at the $1^\circ \times 1^\circ$ resolution. Building a high-density gauge network in a selected area should greatly improve our understanding of the physical processes associated with topographic forcing at the micro- to mesoscales. Additionally, orographic precipitation under different synoptic conditions at various spatial and temporal scales should remain as the focus of future work.

Acknowledgments. This study was in part supported by grants from NASA (NNG05GB85G/EOS03-0063-0069), NSFC (40599424 and 40331006), China MOST (2005CB422006), CMA (GYHY200706209), and the University of San Diego (FRG 06-07 and 07-08). The authors thank the Climate Data Center at the National Meteorological Information Center of the China Meteorological Administration for providing the station precipitation data used in this study. Thanks also to three anonymous reviewers who provided constructive comments and suggestions.

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APPENDIX A

Weather Stations Used in this Study (Source: China Meteorological Administration)

Station no.	Chinese name	Pinyin	Lat (°N)	Lon (°E)	Elev (m)
51709	喀什	Kashi	39.46667	75.98333	1291
51777	若羌	Ruoqiang	39.03333	88.16667	889
51828	和田	Hetian	37.13333	79.93333	1375
51855	且末	Qiemuo	38.15000	85.55000	1248
51886	茫崖	Mangya	38.35000	90.21667	3139
52602	冷湖	Lenghu	38.83333	93.38333	2734
52633	托勒	Tuole	38.80000	98.41667	3368
52645	野牛沟	Yieniugou	38.41667	99.58333	3181
52657	祁连	Qilian	38.18333	100.25000	2789
52707	小灶火	Xiaozaohuo	36.80000	93.68333	2768
52713	大柴旦	Dachaidan	37.85000	95.36667	3174
52737	德令哈	Delingha	37.36667	97.36667	2982
52754	刚察	Gangcha	37.33333	100.13333	3302
52765	门源	Menyuan	37.38333	101.61667	2851
52787	乌鞘岭	Wuqiaoling	37.20000	102.86667	3044
52818	格尔木	Germu	36.41667	94.90000	2809
52825	诺木洪	Nuomuhong	36.43333	96.41667	2791
52833	乌兰	Wulan	36.91667	98.48333	2951
52836	都兰	Dulan	36.30000	98.10000	3192
52842	茶卡	Chaka	36.78333	99.08333	3089
52856	恰卜恰	Qiabuqia	36.26667	100.61667	2836
52866	西宁	Xining	36.61667	101.76667	2262
52868	贵德	Guide	36.03333	101.43333	2238
52876	民和	Minhe	36.31667	102.85000	1815
52908	伍道梁	Wudaoliang	35.21667	93.08333	4614
52943	兴海	Xinghai	35.58333	99.98333	3324
52955	贵南	Guinan	35.58333	100.75000	3202
52957	同德	Tongde	35.26667	100.65000	3290
55228	狮泉河	Shiquanhe	32.50000	80.08333	4279
55248	改则	Gaize	32.15000	84.41667	4416
55279	班戈	Bange	31.38333	90.01667	4701
55294	安多	Anduo	32.35000	91.10000	4801
55299	那曲	Naqu	31.48333	92.06667	4508
55437	普兰	Bulan	30.28333	81.25000	3901
55472	申扎	Shenzha	30.95000	88.63333	4674

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Station no.	Chinese name	Pinyin	Lat (°N)	Lon (°E)	Elev (m)
55493	当雄	Tangxiong	30.48333	91.10000	4201
55569	拉孜	Lazi	29.08333	87.60000	4001
55572	南木林	Nanmulin	29.68333	89.10000	4001
55578	日喀则	Rikaze	29.25000	88.88333	3837
55589	贡嘎	Gongge	29.30000	90.98333	3556
55591	拉萨	Lhasa	29.71667	91.03333	3659
55593	墨竹贡卡	Muozhugongka	29.85000	91.73333	3806
55598	泽当	Zedang	29.25000	91.76667	3553
55655	聂拉木	Nielamu	28.18333	85.96667	3811
55664	定日	Dingri	28.63333	87.08333	4302
55680	江孜	Jiangzi	28.91667	89.60000	4041
55681	浪卡子	Langkazi	28.96667	90.40000	4433
55696	隆子	Longzi	28.41667	92.46667	3861
55773	帕里	Pali	27.73333	89.08333	4301
56004	托托河	Tuotuohe	34.21667	92.43333	4534
56016	治多	Zhiduo	33.85000	95.60000	4181
56018	杂多	Zaduo	32.90000	95.30000	4069
56021	曲麻莱	Qumalai	34.55000	95.48333	4232
56029	玉树	Yushu	33.01667	97.01667	3682
56033	玛多	Maduo	34.91667	98.21667	4273
56034	清水河	Qingshuihe	33.80000	97.13333	4418
56038	石渠	Shiqu	32.98333	98.10000	4201
56046	达日	Dari	33.75000	99.65000	3969
56065	河南	Henan	34.73333	101.60000	3501
56067	久治	Jiuzhi	33.43333	101.48333	3630
56079	若尔盖	Ruoergan	33.58333	102.96667	3441
56080	合作	Hezuo	35.00000	102.90000	2911
56093	岷县	Minxian	34.43333	104.01667	2316
56106	索县	Suoxian	31.88333	93.78333	4024
56109	比如	Biru	31.48333	93.78333	3941
56116	丁青	Dingqing	31.41667	95.60000	3874
56125	囊谦	Rangqian	32.20000	96.48333	3645
56137	昌都	Changdu	31.15000	97.16667	3307
56144	德格	Dege	31.73333	98.56667	3199
56146	甘孜	Ganzi	31.61667	100.00000	3394
56151	班玛	Banma	32.93333	100.75000	3530
56152	色达	Ceda	32.28333	100.33333	3896
56167	道孚	Daofu	30.98333	101.11667	2959

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Station no.	Chinese name	Pinyin	Lat (°N)	Lon (°E)	Elev (m)
56172	马尔康	Markang	31.90000	102.23333	2666
56173	红原	Hongyuan	32.80000	102.55000	3493
56178	小金	Xiaojin	31.00000	102.35000	2369
56182	松潘	Songpan	32.65000	103.56667	2852
56202	嘉黎	Jiali	30.66667	93.28333	4490
56223	洛隆	Luolong	30.75000	95.83333	3640
56227	波密	Pomi	29.86667	95.76667	2737
56247	巴塘	Batang	30.00000	99.10000	2589
56251	新龙	Xinlong	30.93333	100.31667	2999
56257	理塘	Litang	30.00000	100.26667	3951
56287	雅安	Yaan	29.98333	103.00000	629
56312	林芝	Linzhi	29.56667	94.46667	3001
56317	米林	Milin	29.21667	94.21667	2952
56331	左贡	Zuogong	29.66667	97.83333	3781
56342	茫康	Mangkang	29.68333	98.60000	3871
56357	稻城	Daocheng	29.05000	100.30000	3729
56374	康定	Kangding	30.05000	101.96667	2616
56434	察隅	Chayu	28.65000	97.46667	2331
56444	德钦	Deqin	28.45000	98.88333	3488
56462	九龙	Jiulong	29.00000	101.50000	2994
56548	维西	Wixi	27.16667	99.28333	2322

APPENDIX B

Rotated PC Scores Representing Terrain and Fetch Variables

Station ID	RPC1	RPC2	RPC3	RPC4	RPC5	RPC6	RPC7	RPC8
51709	−0.315	−2.442	−0.892	2.603	3.816	1.340	2.348	0.464
51777	0.145	−2.729	−1.478	−1.271	−1.572	1.117	−2.769	0.115
51828	−0.298	−2.002	−1.701	−1.608	−0.880	2.102	2.293	0.385
51855	−1.106	−2.434	−1.530	−1.544	−2.581	0.313	−2.728	0.429
51886	0.208	−0.341	−2.573	2.031	1.431	−0.380	−0.216	−0.359
52602	0.589	−1.013	2.291	2.353	−0.873	1.993	−0.906	−0.692
52633	0.554	−0.038	0.577	0.827	−1.786	−1.170	1.307	−0.602
52645	0.302	−0.656	−0.898	0.550	−0.563	−2.055	0.768	−0.252
52657	0.640	−0.566	−0.471	0.485	−1.379	−1.597	0.924	−0.281
52707	−0.618	−0.230	−1.954	−1.532	−1.073	2.501	4.765	−0.322
52713	0.287	−0.282	2.117	1.267	−0.646	1.516	−1.030	−0.936
52737	−0.135	−0.490	0.499	3.233	−1.118	0.769	−1.131	−1.008
52754	−0.583	−0.523	1.457	0.959	1.000	0.057	0.080	0.282
52765	0.535	−0.614	0.389	1.152	−0.636	−0.835	0.511	−0.296
52787	0.442	−1.269	−0.818	0.365	−0.275	−1.133	0.211	−0.315
52818	−0.082	−0.343	−2.832	−1.257	−2.548	0.940	0.050	−0.395
52825	0.114	−0.552	−1.471	0.043	−2.076	1.498	0.183	−0.736
52833	−0.269	−1.081	0.632	2.228	−1.519	−1.420	−0.250	0.587
52836	−0.367	−0.430	1.741	−0.091	−1.825	1.701	−1.645	−1.050
52842	−0.505	−0.628	1.380	0.719	−1.252	−0.901	1.848	−0.708

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Station ID	RPC1	RPC2	RPC3	RPC4	RPC5	RPC6	RPC7	RPC8
52856	−0.219	−0.954	0.395	1.530	−0.432	−0.002	0.070	−0.491
52866	−0.118	−1.504	0.279	0.363	0.697	−0.564	0.144	−0.304
52868	0.287	−0.985	0.506	0.016	−0.872	−0.559	0.379	−0.515
52876	−0.194	−1.875	0.387	0.273	1.770	0.636	−0.022	−0.773
52908	−1.084	1.562	−1.908	2.281	−0.322	0.729	1.691	−1.034
52943	−0.508	−0.447	−0.044	−0.456	0.287	−0.239	1.454	−0.399
52955	−0.663	−0.886	−0.401	0.547	−0.760	0.157	−0.037	0.407
52957	−0.162	−1.253	−0.120	−0.465	−1.435	−1.690	−0.583	2.324
55228	0.163	1.329	0.350	0.095	−0.607	−0.235	0.553	−0.960
55248	−0.860	1.040	−0.174	0.285	−1.148	−0.239	−0.083	−0.198
55279	−0.915	1.160	0.039	0.392	−0.528	0.599	0.089	1.652
55294	−1.144	1.503	−0.396	0.127	0.217	2.195	−1.489	0.662
55299	−1.236	0.795	−0.145	0.419	0.531	0.706	−1.170	0.252
55437	2.560	1.503	−0.997	−0.261	−0.537	0.988	0.614	0.503
55472	−0.962	1.335	−0.682	−0.568	0.590	−0.653	−0.525	−1.425
55493	−0.208	1.551	−1.506	−0.178	0.277	−0.051	−1.019	−0.660
55569	−0.161	1.175	−0.812	0.596	0.315	−0.521	−0.544	−0.955
55572	−0.185	1.029	−0.559	0.835	0.662	−0.049	−0.671	0.660
55578	−0.343	0.169	−0.936	1.267	−0.207	−0.989	−0.207	0.062
55585	−0.297	0.796	−0.293	−0.471	1.123	−0.749	−0.976	−0.957
55589	−0.324	0.184	−0.607	−0.201	0.214	−1.087	−0.454	0.039
55591	−0.075	0.720	−0.315	0.344	0.673	0.381	−0.706	−0.162
55593	−0.158	0.465	−0.306	0.398	−0.200	−0.973	−0.582	1.314
55598	−0.206	0.873	−0.160	−0.184	0.571	−0.388	−0.427	−0.681
55655	3.914	0.903	0.184	0.839	0.305	2.895	−0.569	1.092
55664	−0.157	0.807	−0.724	0.860	0.082	−0.751	−0.274	−0.625
55680	−0.395	0.587	−0.726	−0.350	−0.101	−1.192	−0.441	0.228
55681	0.343	1.261	−1.112	−0.756	0.053	−0.468	−0.302	−0.555
55696	1.113	1.236	−1.356	0.657	0.003	−0.423	−0.597	−0.183
55773	2.617	0.799	−0.492	0.429	0.867	1.121	−0.426	0.604
56004	−1.106	0.683	−0.551	0.652	0.399	0.059	−0.672	3.095
56018	−0.791	0.693	0.890	0.307	0.405	−0.471	0.323	2.367
56021	−1.124	0.225	0.692	−0.793	0.088	0.098	0.453	1.745
56029	−0.562	0.252	−0.240	−0.249	−0.327	−0.742	−0.310	2.652
56033	−1.091	0.581	0.668	1.557	−0.611	1.484	0.730	0.421
56034	−0.463	0.486	0.450	−0.652	0.291	0.163	0.175	4.772
56038	−0.965	0.924	0.502	−1.334	0.755	0.960	1.018	−0.178
56046	−1.104	0.139	1.195	−0.042	−0.481	0.379	1.637	1.385
56065	−1.557	−0.195	0.109	−0.637	1.089	−0.131	−0.609	−0.600
56067	−0.731	0.030	−0.479	−0.200	1.620	1.303	0.042	−1.057
56079	−1.075	−0.431	1.884	−0.799	0.326	1.967	−0.687	−0.675
56080	−0.551	−1.021	−0.060	−0.056	0.844	−0.683	−0.614	−0.827
56093	0.039	−1.415	0.299	0.068	0.238	−1.424	0.653	−0.153
56106	−1.158	0.106	−0.704	1.345	0.336	−0.444	−0.831	1.079
56109	−0.701	0.967	−0.607	−0.750	0.597	−0.490	0.348	−0.431
56116	−0.193	0.812	0.041	0.355	−0.076	−0.494	−0.142	−0.871
56125	−0.773	0.280	1.392	−0.546	−0.648	−0.067	0.392	0.398
56137	−0.407	0.355	1.164	−0.426	0.030	−0.817	0.704	−0.043
56144	0.395	0.926	0.832	−0.863	−0.058	−0.309	0.373	−0.899
56146	−0.055	0.271	0.232	−0.181	−0.271	−1.012	0.532	−0.732
56151	−0.891	0.287	1.331	−0.550	0.191	1.286	0.662	−0.214
56152	−0.676	0.180	1.812	−0.772	0.291	0.129	0.449	1.045
56167	−0.053	0.168	1.047	−1.046	−0.055	−0.590	0.584	−0.853
56172	0.122	−0.151	0.235	−0.874	0.248	−0.430	0.190	−0.323
56173	−1.116	0.304	−0.092	−1.817	0.961	1.306	−2.008	−0.915
56178	1.947	−0.206	0.043	−0.210	0.259	−0.412	−0.197	0.480
56182	0.747	−0.444	0.216	−0.550	0.760	−0.372	−0.186	−0.227
56202	−0.503	1.311	−0.725	−0.028	0.167	−0.650	−0.192	−0.788
56223	0.238	0.919	0.279	−0.247	−0.287	−0.063	0.478	−0.656
56227	2.201	−0.110	0.941	−0.119	−0.285	0.111	0.250	0.381
56247	0.367	0.336	0.794	−1.315	0.395	−0.514	−0.104	−0.720

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Station ID	RPC1	RPC2	RPC3	RPC4	RPC5	RPC6	RPC7	RPC8
56251	−0.532	0.261	0.948	−1.040	0.495	−0.463	−0.124	−0.647
56257	−0.629	0.622	0.379	−1.180	0.315	0.603	−0.260	−0.962
56287	−0.034	−3.457	−1.036	−0.840	3.389	0.131	−0.629	0.104
56312	1.186	0.195	−0.624	−0.379	0.447	−0.700	−0.219	−0.044
56317	1.572	−0.227	−0.290	−0.017	0.565	−0.857	−0.007	0.261
56331	1.115	0.153	1.245	−0.185	−0.934	−0.242	0.423	−0.330
56342	0.225	−0.046	1.173	−1.508	0.613	−0.738	0.148	−0.660
56357	−0.196	0.275	0.988	−0.297	0.332	−0.748	0.243	−0.865
56374	2.569	−0.093	0.248	−0.961	0.214	0.479	0.201	0.653
56434	1.495	0.006	0.329	0.262	0.505	0.414	−1.048	−0.077
56444	2.177	0.039	0.657	−1.470	0.089	−0.236	0.372	0.344
56462	2.044	0.275	−0.318	−0.557	0.628	−0.231	0.113	0.435
56548	0.636	−1.481	0.876	−1.229	0.413	−1.484	−0.154	−0.103

Station ID	RPC9	RPC10	RPC11	RPC12	RPC13	RPC14	RPC15	RPC16
51709	−0.190	3.804	0.106	0.075	1.017	0.375	−0.622	−1.484
51777	−0.524	−0.681	−0.145	0.237	−0.166	−1.075	0.481	−0.377
51828	−0.448	−1.209	−0.308	−1.251	−0.637	0.291	0.034	−0.574
51855	0.139	6.959	−0.052	−0.071	−0.420	−0.685	0.498	1.087
51886	−0.254	−0.498	−0.802	−0.383	−0.796	0.971	−0.474	0.291
52602	−0.415	−0.768	−0.326	0.476	0.157	−1.163	0.972	1.395
52633	0.476	0.230	−0.658	0.286	0.187	0.784	0.041	1.960
52645	−0.224	−0.296	−0.222	0.454	−0.111	1.404	−0.134	1.433
52657	−0.128	−0.242	−0.251	0.751	−0.412	0.000	−0.162	0.911
52707	0.107	0.299	0.342	−0.377	1.051	1.220	−0.152	0.221
52713	0.138	−0.332	−0.669	−0.199	0.226	1.107	−0.423	1.035
52737	−0.244	−0.389	−0.734	0.518	−0.416	−0.910	−0.983	−1.287
52754	−1.136	0.133	0.487	−0.447	0.145	0.638	7.645	−0.203
52765	−0.204	−0.183	0.025	0.176	−0.148	0.209	−0.544	1.891
52787	−0.336	−0.628	−0.852	0.129	0.053	−0.257	−0.245	2.195
52818	0.018	−1.836	−0.220	−0.554	−0.068	−0.170	0.570	−0.411
52825	−0.599	−1.953	−0.580	−1.100	0.268	−0.978	0.055	−1.178
52833	0.823	−0.479	−0.617	−0.630	−1.181	−0.753	−0.912	−0.203
52836	−0.459	−0.506	0.441	−0.471	0.190	0.240	−0.833	−1.839
52842	−0.419	−0.062	0.214	−1.130	0.524	−1.468	0.067	0.724
52856	0.263	−0.107	0.296	0.157	−0.385	0.815	2.010	−0.325
52866	−0.752	−0.842	−0.345	−0.487	0.677	1.304	−0.180	−0.132
52868	−0.360	−0.063	−0.002	0.532	−0.177	0.783	−0.341	−0.520
52876	0.659	−0.712	−0.581	1.086	1.373	−2.107	−0.042	0.564
52908	−1.752	1.709	−0.653	3.294	−1.132	−0.831	−0.394	−1.222
52943	4.284	−0.121	−0.112	−0.110	−0.790	−0.591	−0.456	1.023
52955	0.070	−1.163	3.621	0.634	−0.194	0.039	−0.424	−0.268
52957	0.874	−0.465	−0.604	0.515	1.835	3.496	−0.017	−0.588
55228	0.422	0.580	−1.167	0.043	0.347	1.030	0.093	−0.415
55248	−0.336	−0.218	−0.161	0.526	1.836	−0.704	0.279	0.167
55279	−0.508	−0.640	−0.394	−0.830	0.774	−0.455	−0.483	0.504
55294	−0.998	−0.605	−1.606	1.494	−0.919	1.749	−0.382	−0.633
55299	−0.198	−0.345	−0.173	−0.349	3.520	−0.166	−0.501	0.586
55437	−0.165	0.273	−0.040	0.117	0.239	−0.264	1.567	1.210
55472	0.174	−0.118	−0.536	0.142	0.370	−0.432	1.777	0.906
55493	1.106	−0.333	0.013	0.324	−0.414	0.370	2.365	−0.094
55569	−0.625	0.209	−0.501	−0.848	0.403	−0.145	−0.283	−0.592
55572	2.153	0.210	1.061	−0.846	−0.068	0.823	−0.362	−0.319
55578	−0.483	0.012	−0.878	−0.647	1.170	−0.037	−0.345	−2.193
55585	−0.648	0.045	−0.107	0.636	−0.890	0.877	−0.461	−0.581
55589	2.180	−0.010	−0.706	−1.233	−0.922	0.178	0.537	−1.149
55591	4.309	0.122	−0.080	−1.037	0.091	0.225	−0.542	−0.663
55593	1.162	0.011	−0.970	−0.547	−0.922	0.230	−0.233	−1.539
55598	−0.683	0.267	1.111	−0.568	−0.376	−0.951	−0.144	−0.937
55655	0.029	0.157	−0.134	−0.247	−0.293	−0.646	−0.570	0.457
55664	−0.799	−0.042	−0.013	−0.609	0.756	0.902	−0.111	−0.728

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Station ID	RPC9	RPC10	RPC11	RPC12	RPC13	RPC14	RPC15	RPC16
55680	−0.383	−0.006	1.417	0.446	1.098	−0.162	−0.005	−0.949
55681	0.551	−0.216	−0.244	0.392	−0.089	0.404	1.776	−0.105
55696	−0.299	−0.113	−0.163	−0.371	−0.102	0.276	−0.115	1.718
55773	−0.252	0.381	0.155	−0.338	0.066	−0.052	−0.494	1.968
56004	0.156	0.041	−0.208	0.952	3.047	0.418	−0.158	1.179
56018	−1.219	0.222	−1.124	−3.221	−1.050	−1.246	−0.130	0.233
56021	0.679	−0.733	−1.228	2.842	−1.613	−2.077	0.395	0.251
56029	−0.345	−0.325	1.054	−0.136	0.049	−1.016	−0.436	0.314
56033	3.121	0.489	1.684	−0.796	−2.122	0.035	1.003	0.264
56034	−0.849	−0.403	−1.085	1.371	−0.774	0.344	−0.103	−0.204
56038	−1.006	0.297	−1.214	−2.143	−0.899	−0.428	−0.432	0.385
56046	−0.156	−0.182	0.546	2.512	−1.150	−1.602	0.311	0.251
56065	0.112	−0.648	−0.901	1.988	−1.886	−0.611	−0.858	1.028
56067	0.354	−0.673	−0.105	−0.192	2.268	−0.567	−0.132	0.851
56079	−0.109	0.090	1.826	0.295	−0.532	2.997	−0.620	0.698
56080	−0.839	−0.622	−0.606	0.670	−0.419	1.280	−0.347	1.379
56093	−0.465	−0.313	−0.684	−0.190	−0.345	0.051	−0.163	0.491
56106	−0.271	−0.755	4.488	−0.981	−0.183	−0.294	−0.693	−0.033
56109	−1.365	−0.329	0.520	−0.284	−0.688	−1.630	0.413	−0.355
56116	0.239	0.339	−0.735	0.651	−1.002	0.324	−0.547	−0.725
56125	0.142	0.343	0.317	0.153	2.964	−0.988	−0.349	0.043
56137	−0.853	0.431	0.401	−1.887	−1.030	−1.214	−0.257	−0.768
56144	0.492	0.496	−0.522	0.707	−0.074	0.125	−0.030	0.186
56146	−0.560	0.333	−1.192	−0.968	−0.126	0.647	0.082	−0.386
56151	−0.492	−0.064	0.571	0.803	−0.469	3.586	−0.739	−0.227
56152	−0.714	0.382	−1.691	−2.950	−1.132	0.939	−0.128	0.471
56167	−0.296	0.318	0.639	0.338	−0.251	1.018	−0.277	0.052
56172	−0.905	−0.091	3.013	0.314	−0.437	−0.747	−0.444	−0.643
56173	−0.888	0.441	−0.616	−1.089	−1.011	−0.143	−0.564	1.151
56178	−0.355	0.437	0.387	−0.005	−0.238	0.059	−0.173	−0.116
56182	−0.856	−0.152	1.751	0.201	−0.225	−0.220	−0.442	1.905
56202	−0.715	0.176	2.117	−0.295	−0.398	−0.090	−0.241	0.287
56223	−0.041	0.389	−0.351	−0.552	1.181	−0.618	−0.402	−0.473
56227	0.118	0.350	−0.300	0.500	0.544	−0.724	0.215	−3.231
56247	−0.715	0.233	0.815	0.694	−0.143	0.524	−0.179	−1.289
56251	0.548	0.070	−0.948	1.537	−0.474	−0.500	−0.645	−0.586
56257	2.458	0.003	−0.024	0.428	1.330	−1.245	−0.221	1.100
56287	0.497	−2.107	−0.684	−0.516	−0.588	−0.662	0.322	−1.059
56312	−0.254	−0.233	0.603	0.549	−0.980	0.135	−0.276	0.004
56317	−0.489	0.401	0.076	−0.496	−0.161	−0.150	−0.322	0.336
56331	0.535	0.631	0.278	−0.258	0.094	0.035	−0.277	−1.847
56342	0.007	0.351	−0.590	0.758	2.489	−0.912	−0.002	−0.953
56357	−0.661	0.601	0.136	−1.250	0.199	−1.122	−0.688	0.799
56374	0.427	−0.021	0.627	0.583	−0.289	−0.422	0.186	−0.288
56434	−0.149	−0.031	−0.610	−0.070	0.055	0.239	−0.402	−0.516
56444	0.564	0.266	0.199	0.792	0.135	0.030	0.389	−1.488
56462	−0.343	0.561	0.638	0.181	−0.472	0.540	−0.149	1.092
56548	0.344	−0.229	0.050	0.701	0.458	0.166	0.115	−0.305

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